

Volume of truncated pyramids and cones



1 $\frac{448}{3} \text{ cm}^3$

2 720 cm^3

3 263.89 cm^3

4

a

i $8\pi \text{ units}^3$

ii $27\pi \text{ units}^3$

iii 59.69 units^3

b

i $144\pi \text{ units}^3$

ii $\frac{1024\pi}{3} \text{ units}^3$

iii 619.94 units^3

Surface area of truncated pyramids and cones

5

a 6.32 cm

b 231.7 cm^2

6

a

i $75\pi \text{ units}^2$

ii $27\pi \text{ units}^2$

iii 257.61 units^2

b

i $363\pi \text{ units}^2$

ii $147\pi \text{ units}^2$

iii 1212.65 units^2

c

i $135\pi \text{ units}^2$

ii $60\pi \text{ units}^2$

iii 603.19 units^2

d

i $585\pi \text{ units}^2$

ii $260\pi \text{ units}^2$

iii 2042.04 units^2

Other compound solids

7

a 169.65 cm^3

b 292.17 cm^3

c 221.74 cm^3

d 720.00 cm^3

8

a 173.13 cm^2

b 346.88 cm^2

c 245.04 cm^2

d 322.41 cm^2

- 9 a Break the solid in to more familiar solids (e.g., a cylinder and two cones). Then, draw a net of each solid. Only include faces which are on the surface of the compound solid. Next is to calculate the area of each net. Finally, add up the area of all the nets.
- b 193.07 cm^2

- 10 a 15.0 cm b 916.72 cm^3

Applications

- 11 a $38\,475 \text{ cm}^3$ b 30 cm c $11\,400 \text{ cm}^3$ d $27\,075 \text{ cm}^3$

- 12 43.37 m^3

- 13 a 70.55 km^3 b 0.15 km^3 c 70.40 km^3